

Nama : Derren Roosevelt Wijaya

NPM : 252310029

Kelas : TI – 25 – PA2

Tugas 11

Link Github : https://github.com/wijayaderren2-prog/232510029_LabAlgoritma.git

```
1  #include <iostream>
2  #include <vector>
3  #include <queue>
4  #include <string>
5  #include <utility>
6
7  using namespace std;
8
9  struct Edge {
10     int tujuan;
11     int bobot;
12 };
13
14 int main() {
15
16     // Nama kota
17     string kota[7];
18     kota[0] = "Cibinong";
19     kota[1] = "Cilebut";
20     kota[2] = "Tanah Sareal";
21     kota[3] = "Bogor Tengah";
22     kota[4] = "Bogor Utara";
23     kota[5] = "Baranangsiang";
24     kota[6] = "Tajur";
25
26     // Graph
27     vector<Edge> graph[7];
28
29     Edge e;
30
31     e.tujuan = 1; e.bobot = 4;
32     graph[0].push_back(e);
33
34     e.tujuan = 4; e.bobot = 6;
35     graph[0].push_back(e);
36
37     e.tujuan = 2; e.bobot = 3;
38     graph[1].push_back(e);
39
40     e.tujuan = 3; e.bobot = 2;
41     graph[2].push_back(e);
42
43     e.tujuan = 5; e.bobot = 4;
44     graph[4].push_back(e);
45
46     e.tujuan = 6; e.bobot = 2;
47     graph[5].push_back(e);
48 }
```

```

48
49 e.tujuan = 6; e.bobot = 5;
50 graph[3].push_back(e);
51
52 const int INF = 999999;
53
54 int dist[7];
55 int prev[7];
56 bool visited[7];
57
58 for (int i = 0; i < 7; i++) {
59     dist[i] = INF;
60     prev[i] = -1;
61     visited[i] = false;
62 }
63
64 dist[0] = 0;
65
66 // Algoritma Dijkstra
67 for (int i = 0; i < 7; i++) {
68
69     int u = -1;
70     int min = INF;
71
72     for (int j = 0; j < 7; j++) {
73         if (!visited[j] && dist[j] < min) {
74             min = dist[j];
75             u = j;
76         }
77     }
78
79     if (u == -1)
80         break;
81
82     visited[u] = true;
83
84     for (int j = 0; j < graph[u].size(); j++) {
85         Edge x = graph[u][j];
86
87         if (dist[x.tujuan] > dist[u] + x.bobot) {
88             dist[x.tujuan] = dist[u] + x.bobot;
89             prev[x.tujuan] = u;
90         }
91     }
92
93
94
95     }
96
97 }
98
99 int jalur[10];
100 int jumlah = 0;
101
102 int tujuan = 6;
103
104 while (tujuan != -1) {
105     jalur[jumlah] = tujuan;
106     jumlah++;
107     tujuan = prev[tujuan];
108 }
109
110
111 cout << "==== Jalur Terpendek =====\n\n";
112
113 for (int i = jumlah - 1; i >= 0; i--) {
114     cout << kota[jalur[i]];
115
116     if (i != 0)
117         cout << " -> ";
118 }
119
120
121 cout << endl;
122 cout << "\nTotal Bobot = " << dist[6] << endl;
123
124 return 0;
125
126 }
127

```

```
===== Jalur Terpendek =====
```

```
Cibinong -> Bogor Utara -> Baranangsiang -> Tajur
```

```
Total Bobot = 12
```

```
-----
```

```
Process exited after 0.1887 seconds with return value 0
```

```
Press any key to continue . . . |
```